

Analysis of Customer Shopping Behavior

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1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

```
df.info()

<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Customer ID            3900 non-null   int64   
1   Age                    3900 non-null   int64   
2   Gender                  3900 non-null   str      
3   Item Purchased          3900 non-null   str      
4   Category                3900 non-null   str      
5   Purchase Amount (USD)   3900 non-null   int64   
6   Location                3900 non-null   str      
7   Size                    3900 non-null   str      
8   Color                   3900 non-null   str      
9   Season                  3900 non-null   str      
10  Review Rating           3863 non-null   float64  
11  Subscription Status      3900 non-null   str      
12  Shipping Type            3900 non-null   str      
13  Discount Applied         3900 non-null   str      
14  Promo Code Used          3900 non-null   str      
15  Previous Purchases        3900 non-null   int64   
16  Payment Method           3900 non-null   str      
17  Frequency of Purchases    3900 non-null   str      
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- **Missing Data Handling:** Checked for null values and removed missing value rows (37 rows removed) from the `Review Rating` column using
- `df=df[df["Review Rating"].notnull()]`

```
df.shape
(3863, 18)
```

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

performed structured analysis in MySQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender	revenue
▶	Male	155438
	Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount		customer_id	purchase_amount
▶	2	64		1640	65
	3	73		1643	70
	4	90		1644	77
	7	85		1645	90
	9	97		1647	77
	12	68		1648	78
	13	72		1649	69
	16	81		1650	63
	20	90		1652	80
	22	62		1654	93
	24	88		1656	81
	29	94		1659	66
	32	79		1662	86
	33	67		1667	64
	35	91		1671	73
	37	69		1673	73
	40	60		1674	62
	41	76		1676	90
	43	100			

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased	Average Product Rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.79

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	avg_purchase_amt
▶	Express	60.34
	Standard	58.41

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status	total_customers	avg_spend	total_revenue
▶	Yes	1033	59.42	61377
	No	2830	59.81	169252

6. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	Number of Customers
▶	Loyal	3085
	Returning	695
	New	83

7. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
▶	1	Accessories	Jewelry	170
	2	Accessories	Sunglasses	159
	3	Accessories	Belt	159
	1	Clothing	Blouse	169
	2	Clothing	Shirt	168
	3	Clothing	Pants	168
	1	Footwear	Sandals	159
	2	Footwear	Shoes	147
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	159

8. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status	repeat_buyers
▶	Yes	939
	No	2503

9. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group	total_revenue
▶	Young Adult	61533
	Middle-aged	58380
	Adult	55569
	Senior	55147

5. Dashboard in Power BI

Finally, built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.